Physics-Informed Graph Neural Networks with Uncertainty Quantification: A Pre-Registered Evaluation Protocol for Blue Carbon Credit Pricing

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***Abstract*—Flat, rule-based non-permanence buffers price every credit in a voluntary carbon market the same way regardless of how confident the underlying carbon-stock estimate is, so a well- and a poorly-characterized site are discounted identically. We test whether a physics-informed graph neural network (GNN) combined with conformalized quantile regression (CQR) can produce a per-site confidence signal to complement it, on the Ten Thousand Islands mangrove fringe, southwest Florida (a 10,124-node spatial graph from Sentinel-2, Global Mangrove Watch, and field-calibrated soil organic carbon). Model selection (physics-loss weight, feature set, calibration method) used a pre-registered protocol on validation only, confirmed once on a held-out test block never used for any prior decision. A five-seed robustness check found the physics-informed loss's point-accuracy advantage over a no-physics GNN, reported in an earlier single-seed pilot, did not replicate: on test, the no-physics GNN has the best R^2 of three models (-0.383 vs. -0.430 physics-informed and -0.450 linear). The physics-informed model's predictions correlate most strongly with the imposed physical prior on held-out nodes (0.755 vs. 0.682/0.686), evidence the constraint transfers under spatial shift, not independent physical validation; this gap is real but substantially smaller than the earlier single-seed estimate. Two CQR calibration methods were compared: global achieves 91.1%/96.4% coverage at 90%/95% nominal; a spatially localized (Mondrian) variant, pre-registered from validation-only evidence, undercovers more on test (83.4%, 92.4%). This is a genuine negative result, consistent with spatial-conformal theory requiring more local calibration density than this set supplied. A pricing PoC, built on the better-calibrated global intervals, maps interval width to a multiplier ranging 0.50–1.50 across five test nodes, against one fixed 0.800 multiplier under Plan Vivo's flat 20% buffer. Point-prediction accuracy remains weak throughout; the contribution is a pre-registered protocol that catches both a single-run estimate and a calibration refinement overstating their own reliability, not validated point accuracy or a validated pricing model.**

***Index Terms*—blue carbon, graph neural network, physics-informed learning, conformal prediction, uncertainty quantification, carbon credit pricing**

I. Introduction

Voluntary carbon markets discount mangrove restoration and conservation credits for non-permanence risk (carbon later released via erosion, sea-level rise, or disturbance) using a buffer pool: a fixed percentage of issued credits withheld from sale. Verra's VM0033 scores non-permanence risk via a checklist-based tool, and Plan Vivo V5.0 applies a flat 20% buffer, revisited roughly every ten years [1], [2]. Neither resolves risk at sub-project granularity or conditions the discount on stock-estimate confidence. A restored interior stand and an eroding fringe in the same project get the identical discount (a market inefficiency, not merely an academic gap). Carbon-stock models for this ecosystem and region also report point estimates with no calibrated uncertainty: Radabaugh et al.'s field-plot-calibrated spatial model for our pilot region [3], Sanderman et al.'s global mangrove SOC product, our training label [4], and Simard et al.'s global mangrove biomass product [5] report one value per pixel/site, with no per-prediction confidence for pricing to condition on. Conformal prediction gives valid, distribution-free uncertainty for earth-observation regression [6] and for spatial prediction under autocorrelation-induced exchangeability violations [7]; deep ensembles have been shown in an adjacent climate-downscaling setting [8]. However, neither literature pairs calibrated uncertainty with a physics-informed spatial predictor, couples it to a buffer-pool pricing mechanism, or asks whether a calibration strategy (global versus spatially localized) stays valid under a single region's calibration- density constraints: the narrower gap this paper addresses, not the broader multi-region MRV agenda the wider literature separately identifies [9]. This paper combines a physics-informed graph neural network with distribution-free uncertainty calibration to test whether estimation confidence can complement, not replace, a flat non-permanence discount, following physics-informed neural networks [10] and their graph-based extensions [11], applied to mass-conserving flood forecasting [12], [13] but not, to our knowledge, blue carbon.

The scope is deliberately narrow: one pilot region, one physics constraint (a steady-state mass-balance proxy, not a full temporal flux equation), and one primary UQ method (CQR; deep ensembles and MC dropout are future ablations). Every model-selection decision (physics-loss weight, feature set, calibration method) was pre-registered on a validation split, confirmed once on a spatially distinct test block never used for any prior decision, a two-stage protocol we treat as a methodological contribution (Section III-C). Reliable UQ-driven pricing needs protocols that survive this scrutiny: a naive single-run estimate and a theoretically-motivated calibration refinement can each mislead; this paper's two central findings illustrate both. First, a five-seed check found the physics-informed loss's point-accuracy advantage over a no-physics GNN, seen in an earlier single-seed pilot, did not replicate. What does replicate, at a smaller but real margin, is that its

predictions generalize the imposed physical prior to held-out nodes more strongly than either non-physics model. Second, localized (Mondrian) conformal calibration, competitive with global on validation-internal evidence, underperformed global on the test evaluation that matters, a cautionary result (Section V) for UQ-driven pricing on similarly sparse geospatial data. The contribution is this pre-registered evaluation and its honest limits, not validated point-prediction accuracy or a validated pricing model.

II. Related Work

Physics-informed GNNs. Physics-guided ML integrates scientific structure into learned models [14]; PINNs add a PDE-residual loss [10], Graph Network Simulators encode physics via local, permutation-invariant message passing [11]. The closest prior art is flood forecasting [12], [13], all hydrological; none target mangrove carbon or couple the prediction to a priced instrument via UQ (this paper's delta).

Mangrove carbon stock estimation. The dominant approach is per-pixel tabular regression (random forest, XGBoost) on spectral indices. Simard et al.'s global mangrove biomass product [5] was inaccessible here (NASA Earthdata login required). Radabaugh et al. built a field-plot-calibrated, 30 m carbon-stock model for the southwest Florida coast, one sub-region matching our pilot AOI [3]. Sanderman et al. produced the global, field-calibrated 30 m SOC product [4] we train against. A 105-study review confirms belowground-carbon and valuation research remain under-studied versus aboveground biomass [9]. None uses a graph representation or a physical constraint, or reports a calibrated uncertainty interval (the gap this paper addresses).

Uncertainty quantification. MC dropout [15] is cheap but poorly calibrated under shift. Deep ensembles [16] calibrate better but need multiple full models, including in climate-downscaling [8]. CQR [17] gives a finite-sample, distribution-free coverage guarantee under exchangeability, validated for earth observation [6]; that guarantee is conditional on exchangeability spatial autocorrelation can violate. GeoConformal Prediction [7] addresses this with a spatially localized framework, motivating the Mondrian variant we test (Section III-F).

III. Method

A. Pilot region and data

The pilot AOI is the Ten Thousand Islands / Rookery Bay mangrove fringe, southwest Florida (-81.90° to -81.40° lon., 25.90° to 26.30° lat.), chosen for free access to every needed layer: Sentinel-2 L2A via Microsoft Planetary Computer, Global Mangrove Watch (GMW) v3 extent [18], the Sanderman et al. SOC product [4], and a NOAA CO-OPS tide gauge (station 8725114). Field-calibrated validation numbers came from Radabaugh et al. [3].

B. Graph construction

Native 30 m SOC grid cells were aggregated into 5×5 blocks ($\sim$150 m nodes); Sentinel-2 bands were resampled onto this grid (bilinear) and the GMW mask (nearest-neighbor), so the label was never resampled. A node was kept if its block-mean GMW mangrove fraction exceeded 0.5 and it had a valid SOC value, giving 10,124 nodes and 64,366 directed edges (8-connectivity). Node features were NDVI, raw B04/B08/B11 reflectance, and distance to the tide gauge; the label was mean SOC (t C/ha, 0–100 cm). An initial contiguous-band split gave negative test R^2 for every model family, since the held-out band straddled a real regional carbon-density gradient (Radabaugh et al.'s sub-regional figures range 265.1 ± 43.2 to $409 \pm 104.4\,\mathrm{Mg\,C\,ha^{-1}}$ [3]). We replaced this with an 8×8 checkerboard grid of spatial superblocks ($\sim$6 km each, far larger than the 150 m node spacing), assigned by cumulative node count rather than superblock count, since mangrove nodes concentrate in a thin coastal fringe.

C. Test-set protocol and pre-registered selection

An early version of this evaluation used one fixed split, directly comparing test-set metrics across model variants to decide the paper's central claim, using test data beyond its role as a single untouched confirmation. We corrected this before any experiment below: a fresh validation/test block was carved from superblocks never individually inspected for a decision (only touched via gradient updates); the previously-inspected split was folded into the new training pool, since training on previously-seen data does not leak, but evaluating on it again would. The corrected split is 6,465 train / 1,583 validation / 2,076 test nodes (label means 427.9/421.2/399.4 t/ha, AOI mean 421.0). Every later selection decision (physics-loss weight, feature set, calibration method) was made on validation only, by rules fixed beforehand, confirmed once on test (Section IV): a pre-registered "tested X, selected Y" claim this protocol makes defensible, surfacing a genuine instability a fixed split could not reveal.

D. Architecture and physics-informed loss

All models shared a 2-layer GraphSAGE backbone [19] (hidden dim. 16, dropout 0.3), Adam (lr $= 0.01$, weight decay 10^{-3}), early stopping on validation MSE (patience 30). Baseline and physics-informed variants differed only in loss function, isolating the physics term's effect. The full mass-balance relationship is $dC/dt = \text{Accretion} - \text{Export} - \text{Decomposition}$; a single Sentinel-2 date and no flux-tower data ruled out estimating this directly. We imposed a steady-state prior instead: predicted stock was softly encouraged toward an equal-weighted, min-max-normalized (training data only) combination of an accretion proxy (NDVI) and an export proxy (tide-gauge distance, a positional stand-in for tidal energy absent a per-node hydrodynamic layer). Both proxy components are also raw model features (Section III-B); withholding both from a no-physics baseline reduces its test prior-generalization correlation only modestly (0.682 to 0.654), so this is not narrowly a feature-overlap artifact. A second term penalized negative predictions, a mass-balance corollary, not independent. Both combined into one physics loss, $\mathcal{L}_{\text{phys}} = \lambda \cdot \text{MSE}(\hat{y}, \text{proxy}) + w_{\text{nonneg}} \cdot \text{mean}(\text{ReLU}(-\hat{y}_{\text{raw}})^2)$, added to the standard MSE

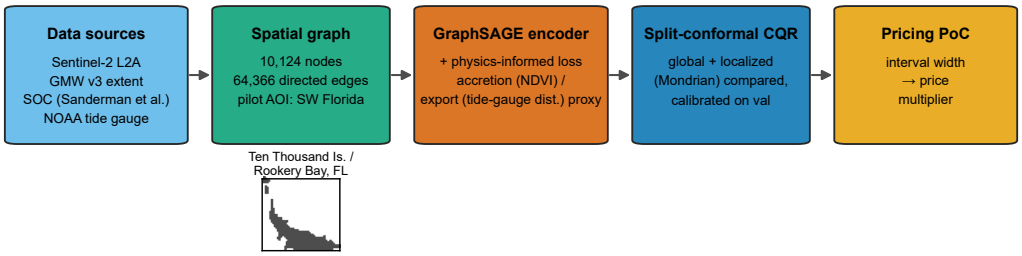


Fig. 1. End-to-end pipeline. Four public data sources feed construction of a 10,124-node, 64,366-edge spatial graph over the pilot AOI (node/edge counts: `src/data_pipeline/build_graph.py` stdout, `data/processed/graph_nodes.csv`; NUMBER_SOURCE_LOG.md #7); a GraphSAGE encoder trained with the added physics-informed loss (Section III-E) feeds a split-conformal CQR head (global and Mondrian variants compared, Section III-F), whose (global) interval widths drive the pricing proof-of-concept (Section VI). Inset: the 10,124 graph node locations (`data/processed/graph_nodes.csv`, columns `lon`/`lat`), the true AOI shape, not a fabricated map.

data loss, with no fitted coefficients inside the proxy itself. $w_{\text{nonneg}} = 0.1$ was held fixed (never swept); λ was chosen by a pre-registered sweep over $\{0.01, 0.05, 0.1, 0.3, 0.5, 1.0, 2.0\}$, five seeds each, on validation only, selecting the value with best mean validation prior-generalization correlation within 0.05 of the best mean validation R^2. This rule selected $\lambda = 0.1$ (mean val. $R^2 = -0.007$, mean val. prior-gen. corr. $= 0.670$), not the ad hoc $\lambda = 0.3$ from an earlier pilot pass, which the rule did not select (full 7-value table in the project's number-source log).

E. Uncertainty quantification: localized CQR

A quantile-head GraphSAGE (same backbone) was trained with pinball loss on the 2.5th, 5th, 95th, and 97.5th percentiles, data loss only (no physics loss; combining both was future work). Two feature sets, NDWI (McFeeters, Green/NIR) and NDMI (Gao, NIR/SWIR), were tested as additions to base, five seeds each, on validation only, under the same tolerance rule as λ; neither improved mean validation prior-gen. correlation within tolerance, so base was retained (Section IV-A).

Split-conformal calibration used the validation set: the conformity score $E_i = \max(q_{\text{lo}}(x_i) - y_i, y_i - q_{\text{hi}}(x_i))$ was computed there, and the $\lceil (n+1)(1-\alpha) \rceil / n$ quantile of $\{E_i\}$ adjusted the interval bounds (*global* calibration, one correction for every node). A *localized (Mondrian)* variant instead partitioned validation nodes by spatial superblock (Section III-B), computing a separate correction per block ($\geq$30 points, else global fallback). Both, plus a density-weighted variant (Gaussian kernel, bandwidth = median nearest-neighbor distance), were compared on a validation-internal holdout (val split 50/50 into calibration/check, never reading test; Section IV-B). Localized (Mondrian) was carried forward to test; global was the comparator, density-weighted a non-adopted check.

IV. Experiments and Results

This section reports measurements only; their interpretation is deferred to Section V.

A. Multi-seed robustness and selection (validation only)

The three-way comparison at the earlier pilot's fixed weight ($\lambda = 0.3$, $w_{\text{nonneg}} = 0.1$) was re-run across five seeds $\{1, \ldots, 5\}$ on validation data. Fig. 2(a) plots mean $\pm$ one std. dev. for validation R^2 and prior-gen. correlation. Baseline GNN: R^2 mean $= 0.005$, std $= 0.019$; corr. mean $= 0.544$, std $= 0.056$. Physics-informed GNN: R^2 mean $= -0.040$, std $= 0.026$; corr. mean $= 0.632$, std $= 0.137$. Linear regression is deterministic: $R^2 = -0.030$, corr. $= 0.486$ (no error bar; no seed dependence to show). The λ sweep and NDWI/NDMI ablation (Section III-E/F) selected $\lambda = 0.1$ and the base feature set mechanically, by the rules stated where each is defined.

B. Localized vs. global CQR (validation-internal, pre-test)

On the validation-internal calibration/check split (Section III-F), at 90 and 95% nominal respectively: global calibration reached 88.0 and 95.6% empirical coverage (mean width 307.6 and 367.8 t/ha); localized (Mondrian) reached 92.2 and 95.6% (width 305.3 and 358.2 t/ha); density-weighted reached 81.1 and 84.7% (width 220.2 and 241.5 t/ha), narrower but visibly undercovering both levels.

C. Final test evaluation (evaluated once)

TABLE I
Model comparison and plausibility metrics, test split (evaluated once)

Model	Test R^2	RMSE	Edge inc.	Prior-gen.
Linear regr.	-0.450	105.4	0.00199	0.686
Baseline GNN	-0.383	103.0	0.0	0.682
Physics GNN	-0.430	104.7	0.0	**0.755**

Table I reports the single test evaluation (train R^2: linear 0.236, baseline GNN 0.276, physics GNN 0.289; negative-prediction rate 0.0 for all three, not shown). The train ranking (physics $>$ baseline $>$ linear) reversed on test: the no-physics baseline GNN had the best test R^2. Edge incoherence (neighboring test-node jumps exceeding the 95th pct. of real training-side neighbor differences, threshold 123.12 t/ha) was 0.0 for both GNNs, near-zero for linear (graph message-passing generically, not the physics term).

TABLE II
CQR COVERAGE, TEST SPLIT (EVALUATED ONCE), GLOBAL VS. LOCALIZED

Nominal	Method	Emp. coverage	Width (t/ha)
90%	Global	91.1%	335.6
90%	Local (Mondrian)	83.4%	317.5
95%	Global	96.4%	382.6
95%	Local (Mondrian)	92.4%	358.8

Table II reports both methods on the single test evaluation. Global met or exceeded nominal coverage at both levels; localized (Mondrian), pre-registered (Section III-F) as the method to carry forward, undercovered more at both levels, despite matching/exceeding global there (Section IV-B). One test node (of 2,076) got an inverted (empty) interval under block-local correction at both levels, floored to zero width and excluded from Section VI's pricing examples; either way it counted as a coverage miss. Fig. 2(b) plots both against the $y = x$ perfect-calibration line.

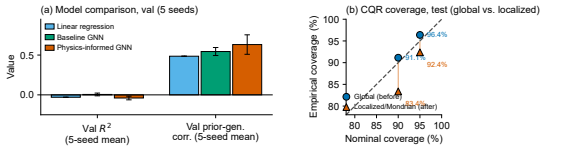


Fig. 2. (a) Validation R^2 and prior-generalization correlation for the three compared models, mean $\pm$ one standard deviation across 5 seeds (Section IV-A; linear regression is deterministic, no error bar). (b) Empirical vs. nominal CQR coverage on the test split, global vs. localized (Mondrian) calibration; dashed line: perfect calibration (Section IV-C).

V. DISCUSSION

The physics term's point-accuracy advantage did not replicate. An earlier single-seed pilot found the physics-informed loss improved test R^2 over a no-physics baseline. The multi-seed check (Section IV-A) already showed this was fragile (physics-informed mean validation R^2 across 5 seeds (-0.040) sat below the baseline's ($+0.005$) even before the corrected split and pre-registered λ), and the final test evaluation confirms it: the no-physics baseline GNN has the best test R^2 of the three. This is the actual finding, not reconciled with the earlier result by reframing; a single seed on a single split was not reliable here.

What does hold up, at a smaller margin. The physics-informed model's predictions correlate with the imposed mass-balance proxy more strongly than either non-physics model on every evaluation in this paper, multi-seed validation (0.632 vs. 0.544/0.486) and the single test evaluation (0.755 vs. 0.682/0.686). This requires the same precise interpretation as before: the proxy correlated against is the exact quantity the physics loss trained toward, so this is a *prior-generalization* check (whether the constraint transfers to held-out nodes), not independent physical validation, which this AOI's single-date, flux-tower-free data cannot supply. Whether this correlation reflects genuine physical behavior, not just successful optimization against its own training target, is this paper's most significant open question. Resolving it would require flux-tower or repeated field-plot carbon-flux measurements at this AOI, which do not currently exist, the clearest concrete next step for extending this work beyond a proxy-based prior. What changed under the corrected protocol is the effect's size, not its direction: a real but modest margin on test (0.069–0.073), not the 0.6-point gap the single-seed pilot reported. This is a direct product of the pre-registered protocol working as intended.

Localized calibration did not improve on global, and that is itself informative. Section III-F pre-registered localized (Mondrian) CQR from validation-internal evidence, before test was read. On test it undercovered more than global at both levels (83.4% vs. 90% nominal; 92.4% vs. 95%, against global's 91.1%/96.4%), likely because only 6 spatial superblocks had enough validation nodes (≥ 30) for a stable local correction. This matches theory: spatial conformal validity requires an *infill asymptotic* regime with sufficient calibration-point density per local region [20], plausibly what our split validation set lacked, a density limit, not evidence against localization. Nor is the pattern isolated to this paper: applying Mondrian calibration to real-estate valuation, Hjort et al. [21] report "little to no improvement" over global in one scenario, cautioning that an apparent gain can instead reflect a non-conformity score failing to capture spatial trends. Localization gains in asset-pricing conformal calibration are setting-dependent, not automatic. **Business implication:** global (simpler, the only method meeting nominal coverage on test) is defensible for near-term deployment; Section VI's pricing PoC uses global intervals accordingly. Both methods still produce test-set widths (318–383 t/ha) several times the label's std. dev. (87.6 t/ha), wide, not falsely confident.

VI. PRICING PROOF-OF-CONCEPT

CQR interval width measures present-day estimation confidence, not non-permanence risk (future carbon loss to erosion, disturbance, sea-level rise) – the quantity buffer pools price; this paper does not establish the two correlate. We illustrate, without validating, how per-node estimation confidence could sit alongside, not replace, a non-permanence buffer, built on *global* CQR intervals per Section V's recommendation, not the localized (Mondrian) variant it cautions against. A relative interval width is the 90%-CQR (global) width divided by the test set's own median (341.1 t/ha; range 143.9–552.5 t/ha). A price multiplier is $1 - \alpha(\text{relative width} - 1)$, clipped to $[0.5, 1.5]$, with $\alpha = 1.0$ an arbitrary illustrative sensitivity, not fit to market outcome. Table III shows five nodes against Plan Vivo V5.0's flat 20% buffer.

TABLE III
PRICING PROOF-OF-CONCEPT: 5 REPRESENTATIVE TEST NODES (GLOBAL CQR)

Node	Interval width	Our mult.	Flat-buffer mult.
Min width	143.9	1.500	0.800
p25	298.5	1.125	0.800
Median	341.1	1.000	0.800
p75	369.5	0.917	0.800
Max width	552.5	0.500	0.800

Concretely, an issuer selling from both the well- (min-width) and poorly-characterized (max-width) node above applies the

identical 20% discount today; under a confidence-adjusted mechanism, the former prices at a 50% premium, the latter at a 50% discount, a plausible implication, not a validated outcome. α and the clipping bounds are stated, arbitrary choices. The flat-buffer value is not arbitrary: it is Plan Vivo V5.0's stated buffer, used only as a comparison point (VM0033 instead scores a project-specific value via its own risk tool, Section I).

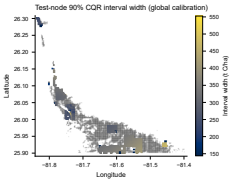


Fig. 3. Per-node 90% global-CQR interval width, 2,076 test nodes (`results/phase2_cqr_test_predictions.csv`, columns `lo_adj_global_0.9`/`hi_adj_global_0.9`; range 143.9–552.5 t/ha, median 341.1 t/ha, matching Table III). Gray points: remaining train/val nodes, AOI context only (Fig. 1's technique). Width clusters by location, narrower northwest and centrally, wider along part of the southern fringe, the visual basis for location-differentiated pricing a flat buffer cannot supply.

VII. Limitations

Single-date imagery constrains the physics term to a cross-sectional relationship, not an actual dC/dt (not invalidating the prior-generalization result, defined relative to that same proxy). **No flux/decomposition data** exists for this AOI; accretion/export are proxies (Section III-E). **Distance-to-gauge is a positional proxy**, not a tidal-energy layer. **The prior-generalization result is not independent validation** (Sections III-F, V): it measures transfer, not physical correctness. **CQR excluded the physics loss**, a disclosed scope choice. **Localized (Mondrian) CQR undercovered on test** (Section IV-C, V), a data-density limit here, not spatial calibration generally [7], [20]. **Single region**; the five-seed check addresses seed variance, not regional generalization. **The pricing mechanism is illustrative**, with an unfit sensitivity parameter and no market validation. **The graph has 2,520 directed edges (3.92%) crossing split boundaries** (9.7% of nodes, 14.2% of test nodes, are 1-hop neighbors across a split); restricting Table I to the 1,782 non-boundary test nodes gives lower R^2 for all three models (linear -0.504, baseline GNN -0.447, physics GNN -0.494) but the same ranking, so reported numbers carry a small, non-model-specific optimistic bias from this leakage. **The single test-side seed, the untested random-forest baseline, and the pricing mechanism's gaming-risk and registry-adoption questions** remain unaddressed.

VIII. Conclusion

A pre-registered, two-stage protocol (validation-only selection, single held-out test confirmation) caught two overstated results a single run would have missed: an earlier single-seed pilot's headline finding, and a theoretically-motivated localized calibration refinement favorable on validation-internal evidence alone. What survives (the physics-informed model's smaller but real prior-generalization advantage, and honestly wide, node-differentiated CQR intervals) are two separate results, not one flowing from the other: the CQR pipeline priced here used data loss only, without the physics term. Together they motivate the market-facing contribution: an estimation-confidence signal that could complement, not replace, flat buffer pools, deployed here on the simpler, better-calibrated global variant. Future work: a second Sentinel-2 date, a higher-volume or second-region localized-conformal test, deep-ensemble/MC-dropout comparisons, and market-outcome validation of the pricing mechanism against non-permanence risk.

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